

KENYA HIGH 2025 PREDICTION



KCSE TEST EXAM

233/3

CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- Write your **name**, admission number and class.
- Answer **all** the Questions in the spaces provided.
- You are not allowed to start working with the apparatus for the first 15 minutes of the 2 ¼ hours allowed for this paper. This time is to enable you to read the Question paper and make sure you have all the chemicals and apparatus you need.
- All working **must** be clearly shown where necessary.
- Mathematical tables and electronic calculators may be used.

FOR EXAMINER'S USE ONLY

Question	Maximum Score	Candidate's Score
1	19	
2	12	
3	09	
TOTAL	40	

Question 1

You are provided with the following:

- Solution **A** ; Hydrochloric acid
- Solution **B**; 0.03M sodium hydroxide
- Solution **C**, Containing 15.74g of $\text{Na}_2\text{CO}_3 \cdot \text{XH}_2\text{O}$ in 250ml of the solution.

You are required to determine: -

- i) The concentration of solution **A**
- ii) The value of **X** in the carbonate $\text{Na}_2\text{CO}_3 \cdot \text{XH}_2\text{O}$

PROCEDURE A

- Fill the burette with solution **A**. Using clean pipette, place 25.0cm^3 of solution **B** into a 250ml conical flask.
- Add 2 drops of **phenolphthalein indicator** and titrate with solution **A**. Record your results in **table I** below. Repeat the experiment two more times and complete the table.

Table I

(4mks)

	I	II	III
Final burette reading (cm^3)			
Initial burette reading (cm^3)			
Volume of solution A (cm^3) used			

a) Determine the :-

(i) Average volume of solution A used.

(1mk)

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(ii) Number of moles of sodium hydroxide in 25 cm^3 of solution **B** used. (1mk)

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(iii) Number of moles of acid in volume of solution **A** used. (1mk)

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(iv) Concentration of solution **A** in moles per litre.

(1mk)

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PROCEDURE B

- Using a **clean** pipette, place 25.0 cm³ of solution **C** into a 250ml volumetric flask. Add about 100cm³ of distilled water. Shake well and add more distilled water to make up to the mark. Label this solution **D**
- Fill the burette with solution **A**. Using a **clean** pipette, place 25 cm³ of solution **D** into a conical flask. Add 2 drops of **methyl orange indicator** and titrate with solution **A**. Record your results in the **table II**.
- Repeat the titration two more times and complete **table II**.

Table II

(4mks)

	I	II	III
Final burette reading (cm ³)			
Initial burette reading (cm ³)			
Volume of solution A (cm ³) used			

b) Determine the: -

(i) Average volume of solution **A** used.

(1mk)

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(ii) Moles of the acid of solution **A** that reacted with the carbonate solution **D**. (1mk)

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(iii) Number of moles of the carbonate in 25 cm³ of solution **D** used. (1mk)

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(iv) Number of moles of carbonate in 250cm³ of solution **D** (1mk)

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(v) Concentration of carbonate solution **C** in **moles per litre**. (1mk)

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(vi) Concentration of the carbonate solution **C** in **grams per litre**. (1mk)

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(vii) Value of **X** in Na₂CO₃. **X**H₂O (H=1.0, C=12.0, O=16.0 Na=23.0) (1mk)

QUESTION 2

You are provided with solid **M**. Use it to carry out the tests below. Write the observations and inferences in the space provided.

(i) Place solid **M** in a boiling tube and add about 10cm³ of distilled water. Divide the resulting solution into **five portions**

Observation	Inference
(1mk)	(1mk)

(ii) To the first portion add sodium hydroxide solution dropwise till excess.

Observation	Inference
(1mk)	(1mk)

(iii) To the second portion add aqueous ammonia solution dropwise till excess.

Observation	Inference
(1mk)	(1mk)

(iv) To the third portion add 3 drops of sodium chloride solution.

Observation	Inference
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(1mk)	(1mk)
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(v) To the fourth portion add about 2cm³ of HNO₃ acid.

Observation	Inference
(1mk)	(1mk)

(vi) To the fifth portion add 3 drops of lead (II) nitrate solution and warm.

Observation	Inference
(1mk)	(1mk)

Question 3

You are provided with solid **G**. Use it to carry out the tests below. Write the observations and inferences in the space provided.

(a) Using a **clean** metallic spatula, take one third of solid **G** and place on Bunsen burner flame.

Observation	Inference
(1mk)	(1mk)

(b) Place the remaining solid **G** in a boiling tube. Add 10cm³ of distilled water and shake the mixture until all the solid dissolves. Divide the resulting solution into **four portions**.

To the first portion add 3 drops of acidified potassium manganate (vii)

Observation	Inference
 (1mk)	 (1mk)

(c) To the second portion add 3drops of bromine water

Observation	Inference
 (1mk)	 (1mk)

(d) To the fourth portion dip universal indicator paper and determine the **pH** of the solution

Observation	Inference
 (1mk)	 (1mk)

(e) To the third portion add solid sodium hydrogen carbonate

Observation	Inference

(½mk)	(½mk)
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